



1/7/2026

ASSIGNMENT DAY 5

PC Assembly and Disassembly.

TASK 1.

❖ List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

➤ Task : Main Components Inside a Desktop PC

<u>Component</u>	<u>Short Description</u>
Motherboard	The motherboard is the main circuit board of the computer. It connects all the hardware components and allows them to work together.
CPU (Central Processing Unit)	The CPU is the brain of the computer. It processes instructions and performs calculations.
CPU Cooler (Heatsink & Fan)	The CPU cooler removes heat from the processor. It keeps the CPU cool and prevents overheating.
RAM (Random Access Memory)	RAM stores temporary data used by running programs. More RAM helps the computer work faster and more smoothly.
GPU (Graphics Processing Unit)	The GPU processes graphics and videos. It improves gaming, video editing, and other graphics-related tasks.
SSD (Solid State Drive)	An SSD stores the operating system and files. It provides fast boot time and quick application loading.
HDD (Hard Disk Drive)	An HDD stores large amounts of data such as documents, photos, videos, and games. It is slower than an SSD.

SMPS (Power Supply Unit)	The SMPS supplies power to all the components inside the computer. It converts AC power into DC power.
Cabinet Fan	Cabinet fans improve airflow inside the PC. They help keep the internal components cool.
CMOS Battery	The CMOS battery powers the BIOS memory. It keeps the system date, time, and BIOS settings saved.
SATA and Power Cables	These cables connect storage devices and provide power to different components for proper operation.

Technical Summary:

- ✓ All these components work together to keep the desktop PC running smoothly and efficiently.

TASK 2.

- ❖ **Open Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step, Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**

➤ Task : Desktop PC Disassembly Process

 **Introduction :** Desktop PC disassembly is the process of removing computer components safely and step by step. It helps us understand the internal parts of a computer and how they are connected.

Desktop PC Disassembly Process



Figure 1: Unplug Power

Turn off the PC and unplug the power cable from the socket.



Figure 2: Remove Side Panel

Remove the screws at the back of the cabinet. Slide or lift the side panel and remove it gently.



Figure 3: Disconnect Cables

Disconnect all power cables and data cables connected to components like HDD, SSD, GPU, and motherboard.



Figure 4: Remove RAM

Press the clips on both sides of the RAM stick. Gently pull the RAM out of the slot.



Figure 5: Remove HDD / SSD

Unscrew the drive bracket (if any). Disconnect the SATA and power cables. Remove the HDD/SSD from the bay.



Figure 6: Remove Motherboard

Unscrew all motherboard screws. Gently lift the motherboard from the cabinet and remove it carefully.

Figure : Step by step desktop PC disassembly process.

Step-by-Step Process :

- ✓ **Figure 1: Unplug Power** Description: The computer was turned off, and the power cable was unplugged before opening the cabinet. This ensures safety while working on the PC.

- ✓ **Figure 2:** Remove Side Panel Description: The screws were removed, and the side panel was opened carefully to access the internal components.
- ✓ **Figure 3:** Disconnect Cables Description: All power and data cables connected to the motherboard and storage devices were disconnected carefully.
- ✓ **Figure 4:** Remove RAM Description: The clips on both sides of the RAM slot were pressed, and the RAM module was removed gently.
- ✓ **Figure 5:** Remove HDD/SSD Description: The storage drive was disconnected from the cables and removed carefully from its mounting position.
- ✓ **Figure 6:** Remove Motherboard Description: The motherboard screws were removed, and the motherboard was lifted out carefully to avoid any damage.

Technical Summary :

- ✓ The desktop PC was disassembled by following the correct sequence of steps. Each component was removed carefully, making the process safe, organized, and easy to understand.

TASK 3.

- ❖ **Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.**

➤ Task : Desktop PC Reassembly Process

 **Introduction :** PC reassembly is the process of installing all the removed components back into the cabinet in the correct order. Following the proper sequence helps ensure safe installation and smooth system operation.

Desktop PC Reassembly Process



Figure 1: Install Motherboard
Place the motherboard in the case and align it with the mounting holes. Secure it with screws.



Figure 2: Install HDD / SSD
Mount the HDD/SSD in the drive bay and secure it with screws.



Figure 3: Install RAM
Align the RAM stick with the slot and press down firmly until the clips lock it in place.



Figure 4: Connect Power & Data Cables
Connect the 24-pin power cable, 8-pin CPU power cable, SATA data cable, and SATA power cable to their respective ports.



Figure 5: Close Side Panel
Place the side panel back on the case and secure it with screws.



Figure 6: Power On & Verify
Connect the power cable and turn on the PC. Check if the system boots successfully and all components are working.

Figure : Step by Step Desktop PC Reassembly Process.

Step-by-Step Process :

- ✓ **Figure 1 : Install Motherboard :** The motherboard was placed inside the cabinet and secured with screws. It is installed first because all other components are connected to it.
- ✓ **Figure 2 : Install HDD/SDD :** The HDD/SSD was installed and connected with the required cables. It stores the operating system, applications, and user data.

- ✓ **Figure 3 : Install RAM** : The RAM module was inserted into the correct memory slot until it locked in place. RAM helps the computer access data quickly and improves performance.
- ✓ **Figure 4 : Reconnect Power and Data Cables** : All power and data cables were connected carefully. This ensures that every component receives power and communicates properly.
- ✓ **Figure 5 : Close the Side Panel** : The side panel was placed back and secured with screws. It protects the internal components and helps maintain proper airflow.
- ✓ **Figure 6 : Power On and Verify** : The power cable was connected, and the PC was turned on to check that all components were working correctly after reassembly.

Technical Summary :

- ✓ The PC was reassembled by following the correct sequence of steps. Installing components in the proper order helps prevent connection errors, protects the hardware, and ensures the system starts and works correctly.

TASK 4.

- ❖ **Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.**

➤ **Task : Safety Precautions and Tools Required Before PC Assembly/Disassembly.**

 **Introduction :** Before assembling or disassembling a PC, it is important to follow safety precautions and use the correct tools. This helps protect the hardware and makes the work safe and easy.

Safety Precautions Checklist :

<u>Safety Precaution</u>	<u>Importance</u>
Turn off the PC and unplug the power cable	Prevents electric shock and protects the components.
Wear an anti static wrist strap or touch a metal surface	Prevents static electricity from damaging sensitive components.
Work on a clean and dry surface	Reduces the risk of dust, moisture, and accidental damage.
Handle components by their edges	Protects delicate circuits from damage and fingerprints.
Keep screws and small parts organized	Prevents losing parts and makes reassembly easier.
Avoid using excessive force	Protects connectors, slots, and hardware from damage.

Ensure proper lighting and ventilation	Makes it easier to identify components and work safely.
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Tools Required :

<u>Tool</u>	<u>Importance</u>
Phillips Screwdriver	Used to remove and tighten PC screws.
Anti-static Wrist Strap	Protects components from electrostatic discharge (ESD).
Thermal Paste	Improves heat transfer between the CPU and heatsink.
Cable Ties	Keeps cables organized and improves airflow.
Soft Cleaning Brush or Compressed Air	Removes dust from PC components.
Small Container for Screws	Keeps screws safe and organized.
Flashlight	Helps to see small connectors and components inside the cabinet.

Technical Summary :

- ✓ Following these safety precautions and using the correct tools makes PC assembly and disassembly safer, easier, and more efficient. It also helps protect the hardware from accidental damage.